PALM steering

palmgroup

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Preface

- This lecture gives instructions on how to run PALM locally in **interactive mode**, i.e. on your local PC/notebook or on any machine where you are logged in (called **local host**). PALM immediately starts to execute.

- Running PALM in **batch mode** (job-queuing/batch system) on a **local** or **remote** (super) computer/host is not part of this seminar. You can find further documentation about how to run PALM in batch mode via link:

  https://palm.muk.uni-hannover.de/trac/wiki/doc/app/palmrun

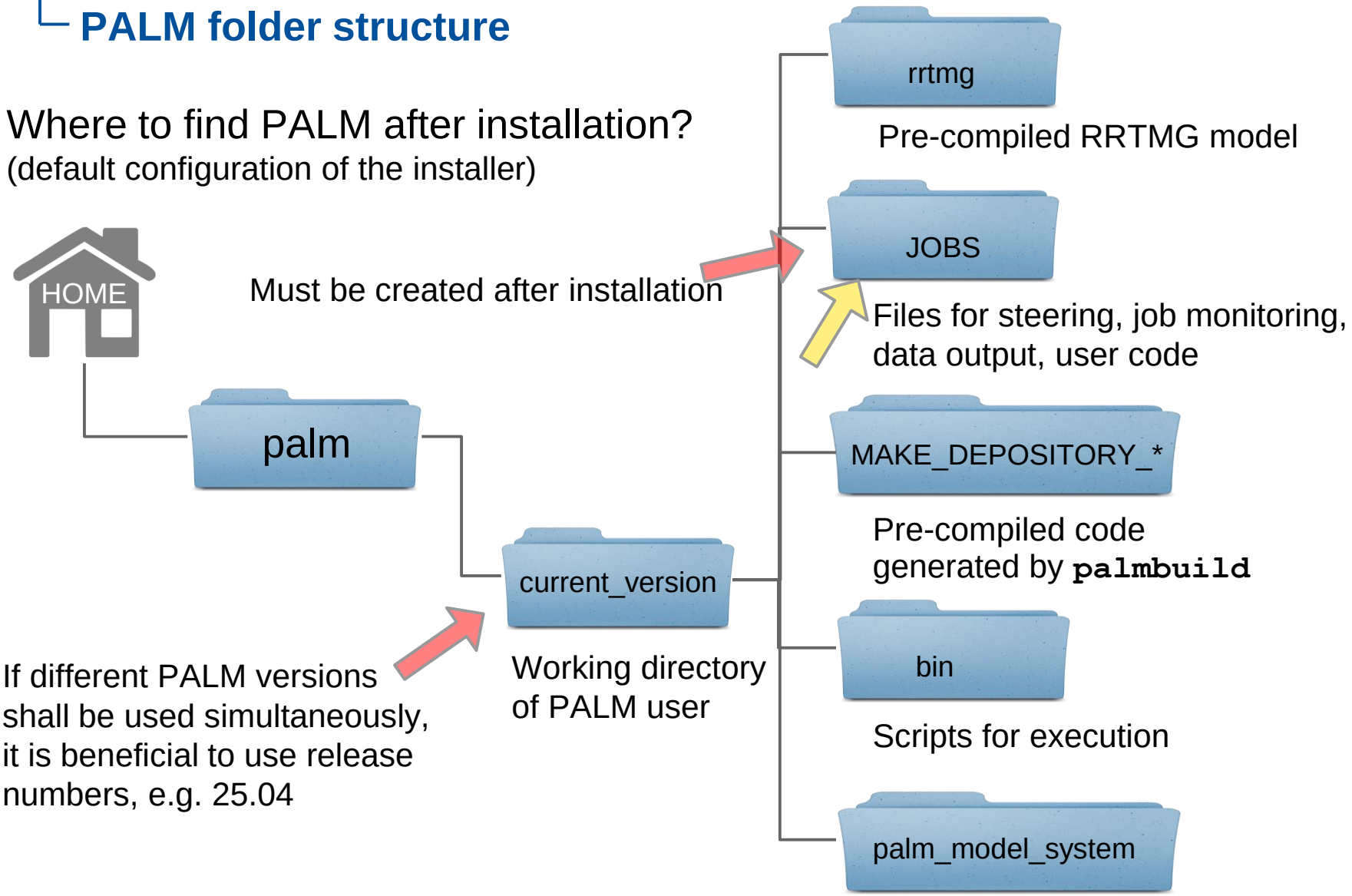
- Information on how to stop and continue a run – the so-called **restart mechanism** – will follow on Wednesday.

Definitions

- **Local host:** The computer that you are currently sitting at or are logged in via your terminal (ssh).
- **Remote host:** Any computer with a batch system, that you have ssh access to, but are not logged in now.
- **Batch mode:** Automatic processing of commands contained in a batch file, not interactively.

PALM folder structure

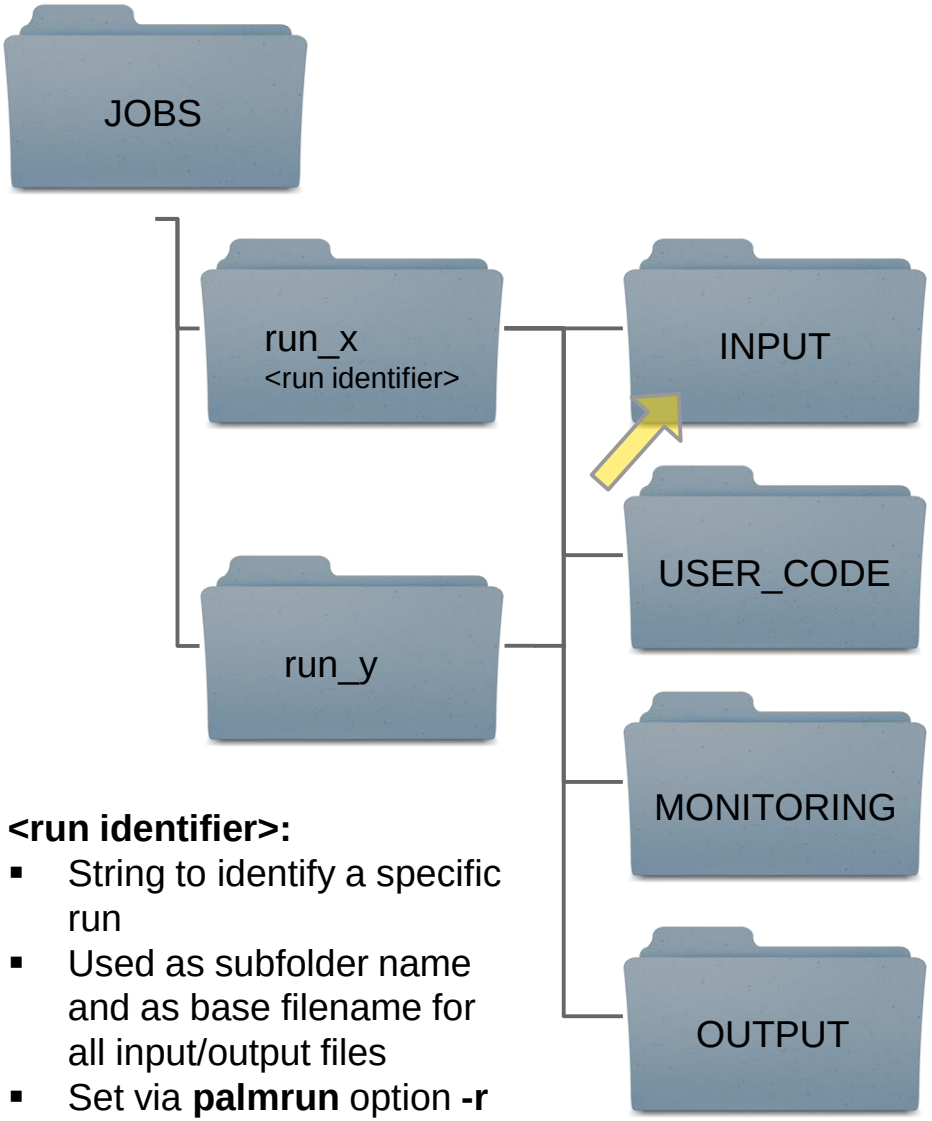
Where to find PALM after installation?
(default configuration of the installer)



Download folder (see lecture „PALM Overview“)

PALM steering

PALM input/output overview



<run identifier>:

- String to identify a specific run
- Used as subfolder name and as base filename for all input/output files
- Set via **palmrun** option **-r**

ASCII-Input | ASCII-Output | NetCDF

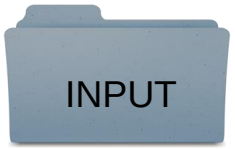
- Parameter file(s) for steering
- Simple topography data
- Driver files (e.g. topography data, soil properties, see lecture „static & dynamic drivers“)

- User-defined source code (see lecture „User-defined code“)

- Header file (parameter settings)
- Run control file (parameter settings, timestep information)
- CPU file (computing time performance measurements)

- 1D profiles xy- & time-averaged)
- 2D sections & 3D data (instantaneous & time-averaged)

Model steering – Parameters overview



- Grid parameters (computational grid / processor grid):
  - grid spacing, number of grid points, etc.
- Numerical parameters & boundary conditions:
  - timestep scheme, advection scheme, etc.
  - horizontal (cyclic/non-cyclic),
  - vertical (constant-flux layer, free-slip condition, etc.)
- Initialization parameters:
  - initial profiles, restart run, main run (following pre-run)
- Output parameters:
  - 1d, 2d, 3d output, output intervals, etc.
- Physical parameters:
  - latitude, angular velocity of the earth, etc.
- Parameters for embedded modules (wind turbine model, land surface model, Lagrangian particle model, etc.)

PALM steering

Model steering – Input parameter file

INPUT

run_x_p3d (<run identifier>_p3d)

```
&initiaization_parameters nx = 39, ny = 39, nz = 40,
        dx = 50.0, dy = 50.0, dz = 50.0,
        dz_stretch_level = 1200.0,

        fft_method = 'temperton-algorithm',

        initializing_actions = 'set_constant_profiles',
        ug_surface = 0.0, vg_surface = 0.0,
        pt_vertical_gradient        = 0.0, 1.0,
        pt_vertical_gradient_level = 0.0, 800.0,
        surface_heatflux = 0.1, bc_pt_b = 'neumann',

        topography = 'read_from_file', /

&runtime_parameters end_time = 3600.0,

        create_disturbances = .T.,
        dt_disturb = 150.0, disturbance_energy_limit = 0.01,

        dt_run_control = 0.0,

        data_output = 'w_xy', 'w_xz', 'w_xz_av', 'theta_xy', ,theta_xz',

        dt_data_output = 900.0,
        dt_data_output_av = 1800.0,
        averaging_interval = 900.0,
        dt_averaging_input = 10.0,

        section_xy = 2, 10, section_xz = 20,

        data_output_2d_on_each_pe = .F.,

        dt_dopr = 900.0,
        averaging_interval_pr = 600.0, dt_averaging_input_pr = 10.0,
        data_output_pr = '#theta', 'w"theta"', 'w*theta*', 'wtheta',
                         'w*2', ,theta*2', /
```

- Written in FORTRAN-NAMELIST syntax
- Read in by PALM
- Two NAMELIST groups mandatory:

  `&initialization_parameters`

  and

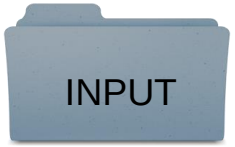
  `&runtime_parameters`
- Further NAMELISTs for specific modules of PALM, e.g.

  `&wind_turbine_parameters`
- Exact formatting required, e.g.:
  - NAMELIST starts with **&<name>** and ends with slash **/**
  - Parameters are separated by **,**
  - Strings are placed between **'...'**
- **Never write comment lines between lists that belong together!**

PALM steering

Model steering – Parameter‘s documentation

https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/



PALM 25.04 (latest) Get Started Guide Reference Search Previous Next Download PALM

- **Namelists**
- Initialization Parameters
- Runtime Parameters
- Agent Parameters
- Biometeorology Parameters
- Bulk Cloud Parameters
- Chemistry Parameters
- Damping Parameters
- Dust Emission and Transport Parameters
- FASTv8 Coupler Parameters
- Flow Control Parameters
- Indoor Parameters
- Land Surface Parameters
- Offline Nesting Parameters
- Nesting Parameters
- Ocean Parameters

Namelists

Initialization Parameters

```
&initialization_parameters
&runtime_parameters
&...
```

advanced_div_correction

Fortran Type: L

Default: *.FALSE.*

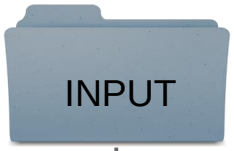
Parameter to enable an improved divergence correction method for the fifth-order scalar advection scheme.

advanced_div_correction = *.T.* activates a better divergence correction for scalar transports with the fifth-order advection scheme. This significantly improves the conservation of scalar quantities. The parameter becomes only effective if scalar_advec = *'ws-scheme'* and requires loop_optimization = *'vector'*.

The fifth-order Wicker-Skamarock scheme is only conditionally conservative, i.e. scalar conservation is achieved as long as the flow divergence is negligible. For topography = *'flat'* this condition is usually guaranteed by the pressure solver. However, in case of topography /= *'flat'* with vertical and elevated horizontal surfaces, the divergence that remains after the pressure solver has been called is not negligible any more at fluid grid points located in the vicinity of walls. There is a default removal of divergence errors implemented in the Wicker-Skamarock scheme, which works sufficiently well in most applications. However, for some cases this default

PALM steering

Model steering – Other input files, e.g.

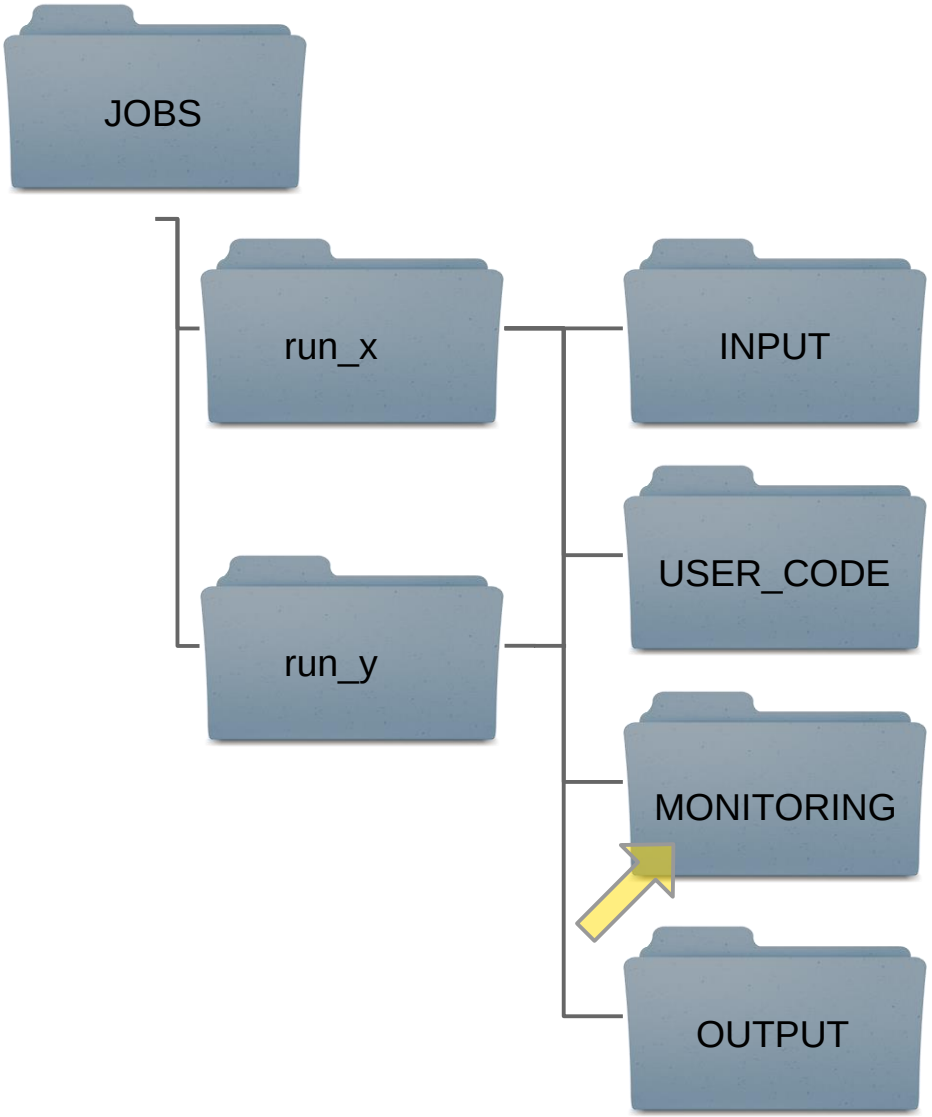


run_x_topo (<run_identifier>_topo)

```
0 0 0 
```

PALM steering

PALM input/output overview



ASCII-Input | ASCII-Output | NetCDF

- Parameter file(s) for steering
- Simple topography data
- Driver files (e.g. topography data, soil properties, see lecture „static & dynamic driver“)

- User-defined source code (see lecture „User-defined code“)

- Header file (parameter settings)
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- 1D profiles xy- & time-averaged)
- 2D sections & 3D data (instantaneous & time-averaged)

PALM steering

Job monitoring – Header

run_x_header (<run identifier>_header.***
with *** = three digit number starting at 000)

```
******************************    ------------------------------------------
* PALM 25.04                 *    atmosphere - run without 1D - prerun
******************************    ------------------------------------------

Date:                  31-07-25    Run:       example_cbl
Time:                  14:56:25    Run-No.:   00
Run on host:              lcmuk
Number of PEs:                4    Processor grid (x,y): (   2,   2) calculated
--------------------------------------------------------------------------------

Numerical Schemes:
-----------------

--> Use the boussinesq approximation for the model equations.
--> Solve perturbation pressure via FFT using temperton-algorithm routines
    perturbation pressure is calculated at every Runge-Kutta step
--> Momentum advection via Wicker-Skamarock-Scheme 5th order
--> Scalar advection via Wicker-Skamarock-Scheme 5th order
--> Loop optimization method: cache
--> Time differencing scheme: runge-kutta-3
--------------------------------------------------------------------------------


Run time and time step information:
----------------------------------

Timestep:              variable     maximum value: 20.000 s    CFL-factor: 0.90
Start time:               0.000 s
End time:              3600.000 s

Time reached:          3607.000 s
CPU-time used:            8.033 s     per timestep:                     0.031 s
                                      per second of simulated time:     0.002 s


Computational grid and domain size:
----------------------------------

Grid length:       dx =     50.000 m    dy =     50.000 m    dz =     50.000 m
Domain size:        x =   2000.000 m     y =   2000.000 m  z(u) =   2862.511 m
```

- Gives general information about the run.
- Summary/Information about the selected model parameters (physical and numerical values).
- File is generated for **every** run (initial as well as restart runs).

PALM steering

Job monitoring – Run control

- For initial runs, the information of the HEADER file is printed at the top of the RUN_CONTROL file.
- File is generated for **every** run, but the HEADER information is only printed for the initial run of a job chain.
- Subsequent to HEADER information, values of specific model variables follow, printed at certain timesteps of the simulation (one line for each timestep, output interval can be controlled by runtime parameter `dt_run_control`).
- **Contents of this timestep output should be carefully viewed after each run, as it allows a first check, if the model ran correctly!**

run_x_rc (<run identifier>_rc)

```
******************************    ------------------------------------------
* PALM 25.04                 *    atmosphere - run without 1D - prerun
******************************    ------------------------------------------

Date:                  31-07-25    Run:       example_cbl
Time:                  14:56:25    Run-No.:   00
Run on host:              lcmuk
Number of PEs:                4    Processor grid (x,y): (   2,   2) calculated
------------------------------------------------------------------------------
...
Run-control output:
------------------

RUN  ITER. HH:MM:SS.SS    DT(E)     UMAX     VMAX     WMAX     U*    W*      THETA*     Z_I     ENERG.   DISTENERG    DIVOLD     DIVNEW     UMAX(KJ
-------------------------------------------------------------------------------------------------------------------------------------------------------
  0      0 00:00:00.00  20.0000A  -0.2131D -0.2288D  0.1182  0.000  1.38   0.000E+00    800.  0.104E-02  0.103E-02  0.597E-03  0.867E-13    7   7
  0      1 00:00:20.00  20.0000A  -0.2125  -0.2273   0.1172  0.009  0.69  -0.131E+02    100.  0.103E-02  0.102E-02  0.570E-05  0.411E-15    7   7
  0      2 00:00:40.00  20.0000A  -0.2117  -0.2252   0.1162  0.009  0.69  -0.132E+02    100.  0.102E-02  0.102E-02  0.939E-05  0.413E-15    7   7
  0      3 00:01:00.00  20.0000A  -0.2106  -0.2227   0.1152  0.009  0.69  -0.132E+02    100.  0.101E-02  0.101E-02  0.131E-04  0.417E-15    7   7
...
```

HEADER

PALM steering

Job monitoring – CPU measurements

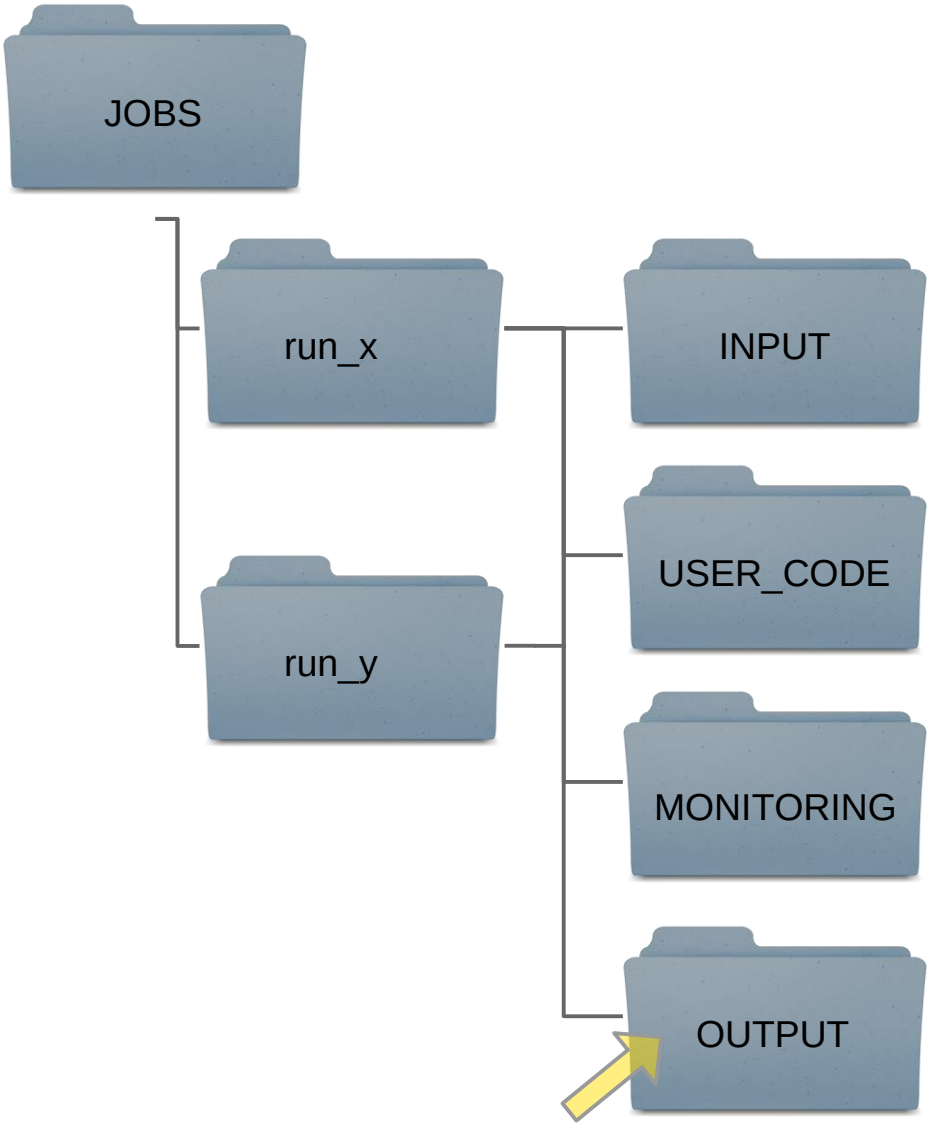
run_x_cpu (<run identifier>_cpu .***
with *** = three digit number starting at 000)

```
PALM 25.04   run: example_cbl.00  host: lcmuk  31-07-25 14:56:25
-----------

CPU measures for     4 PEs (    2(x) *     2(y) tasks *    1 threads):

gridpoints (x/y/z):                        40 *    40 *    40
nr of timesteps:                          263
cpu time per grid point and timestep:       0.45511 * 10**-6 s
----------------------------------------------------------------------

place:                              mean        counts      min       max       rms
                                  sec.      %               sec.      sec.      sec.
-----------------------------------------------------------------------------------
total                             8.043   100.00        1   8.043     8.043     0.000
all progn.equations               5.177    64.37      789   5.054     5.462     0.167
pres                              1.153    14.33      790   1.152     1.154     0.001
exchange-horiz-progn              0.607     7.55      789   0.304     0.738     0.177
surface_layer_fluxes              0.358     4.45      789   0.351     0.364     0.005
diffusivities                     0.322     4.00      789   0.316     0.328     0.004
flow_statistics                   0.239     2.97      264   0.239     0.240     0.001
calculate_timestep                0.059     0.73      264   0.051     0.062     0.005
sum_up_3d_data                    0.013     0.16      147   0.013     0.013     0.000
initialisation                    0.008     0.10        1   0.008     0.008     0.000
data_output_tseries               0.003     0.04      263   0.000     0.000     0.005
disturb_field                     0.003     0.04       10   0.003     0.003     0.000
data_output_2d                    0.002     0.03       10   0.002     0.002     0.000
run_control                       0.002     0.02      264   0.000     0.000     0.003
swap_timelevel                    0.001     0.01      789   0.001     0.001     0.000
last actions                      0.000     0.00        1   0.000     0.000     0.000
user_actions                      0.000     0.00      789   0.000     0.000     0.000
data_output_profiles              0.000     0.00        4   0.000     0.000     0.000
average_3d_data                   0.000     0.00        2   0.000     0.000     0.000

special measures:
-----------------------------------------------------------------------------------
timesteps                         8.033    99.87      263   8.033     8.033     0.000
exchange_horiz                    0.784     9.75     7136   0.483     0.918     0.177
poisfft                           0.647     8.04      790   0.646     0.647     0.001
fft_x                             0.187     2.33      790   0.187     0.188     0.000
fft_y                             0.187     2.33      790   0.186     0.188     0.001
transpo forward                   0.140     1.74      790   0.139     0.142     0.001
divergence                        0.137     1.70     1054   0.136     0.137     0.000
mpi_alltoall                      0.102     1.26     4740   0.101     0.102     0.000
transpo invers                    0.080     0.99      790   0.078     0.080     0.001
tridia                            0.051     0.63      790   0.050     0.051     0.000
```

- Contains information about the CPU requirements of single parts of the program.
- **all progn. equations** and **pres solver** should be the main consumers.
- For larger grids (1024^3 points), **pres** may need up to 50% of the total time.
- Time needed for communication (sum of **exchange_horiz** and **mpi_alltoall**) should not exceed ~20-30% of the total time.
- The load balance among the processor cores can be inspected via **min**, **max** and **rms** (root-mean-square) values → They may indicate network problems!

PALM steering

PALM input/output overview



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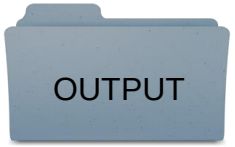
PALM steering

Data output – Available standard output



Type	Possibilities	Local file name generated by PALM run_x_<…>	Filename within temporary working directory and source code
time series	instantaneous	ts.***.nc	DATA_1D_TS_NETCDF
mean vertical profiles	horizontally-averaged & instantaneous or time-av.	pr.***.nc	DATA_1D_PR_NETCDF
2D cross sections (xy, xz, yz)	instantaneous or time-av., possible average along 3rd dimension	xy.***.nc, av_xy.***.nc xz.***.nc, av_xz.***.nc yz.***.nc, av_yz.***.nc	DATA_2D_XY_NETCDF DATA_2D_XY_AV_NETCDF etc.
3D data set	instantaneous or time-av.	3d.***.nc, av_3d.***.nc	DATA_3D_NETCDF DATA_3D_AV_NETCDF
1/2/3D data subsets (masked output)	instantaneous/ time-averaged	masked_M01.***.nc, av_masked_M01.***.nc, masked_M02.***.nc, etc.	DATA_MASK_NETCDF_M01 DATA_MASK_AV_NETCDF_M01 etc.

- NetCDF data format: http://www.unidata.ucar.edu/software/netcdf/
- PALM output steering parameters: https://docs.palm-https://docs.palm-model.org/25.04/Reference/LES_Model/Namelists/#runtime-parameters
- Hints for plotting NetCDF data will be provided in the next lecture „Data visualization“.

PALM steering

PALM folder structure

How to configure PALM I/O and environment?

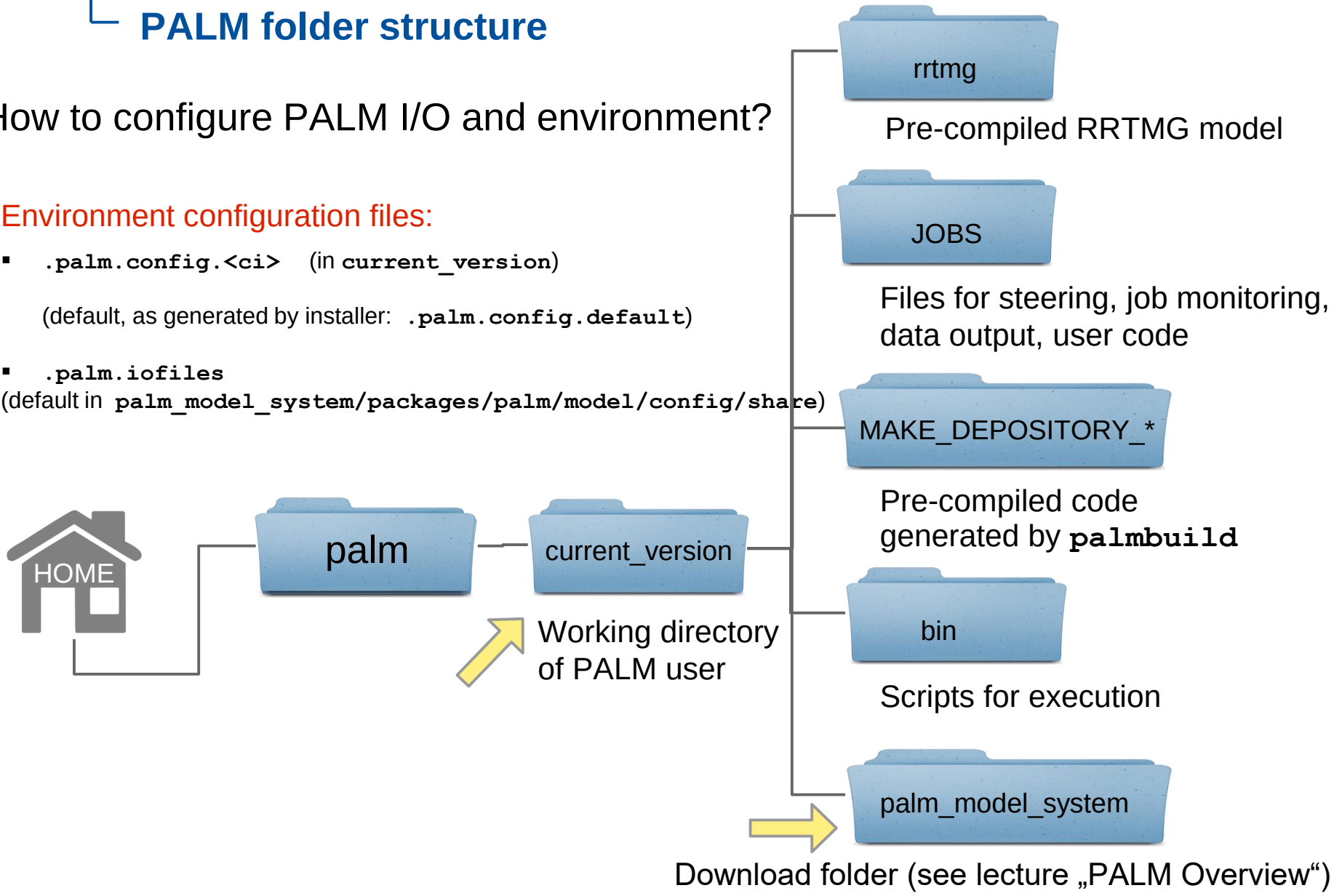
Environment configuration files:

- `.palm.config.<ci>` (in `current_version`)

  (default, as generated by installer: `.palm.config.default`)

- `.palm.iofiles`

(default in `palm_model_system/packages/palm/model/config/share`)



Configuration of Environment – Overview

Which steps are taken care of to run PALM?

palmbuild

(1) Compiles PALM source code
- Creates **`MAKE_DEPOSITORY_<configuration_identifier>`** (short **<ci>**, default: **<ci>** = *default*), containing pre-compiled source code
- Already done, if automatic installer has been used.
- **`palmbuild`** has to be called manually after code or configuration changes!

palmrun

(2) Executes PALM
- **PALM is executed in a temporary folder named** **`<run_identifier>.<randomnumber>`**, which should reside on a fast file system, set via variable **`%fast_io_catalog`** in the configuration file (next slide).

(3) Copies/transfers input & output files
- Input files to **`<ri>.<randomnumber>`**, from input folders, output files to output folders. Cyclic numbers added to output files in order to avoid overwriting in case of running the same/similar jobs multiple times.

(4) By default, the temporary folder **`<ri>.<randomnumber>`** is deleted at the end.

`palmrun` calls **`palmbuild`** (step 1) in case that MAKE_DEPOSITORY_<ci> doesn't exist (new installation) or user-defined code shall be used!

Settings to be used for **`palmrun`** / **`palmbuild`** must be specified by the user in PALM's configuration files (see next slides).

Configuration of Environment – Configuration File

.palm.config.<ci>

```
#
%base_directory      /home/raasch/palm/current_version
%base_data           /home/raasch/palm/current_version/JOBS
%source_path         /home/raasch/palm/current_version/trunk/packages/palm/model/src
%user_source_path    /home/raasch/palm/current_version/JOBS/$run_identifier/USER_CODE
%fast_io_catalog     /localdata/raasch
%restart_data_path   /localdata/raasch
%output_data_path    /home/raasch/palm/current_version/JOBS
%local_jobcatalog    /home/raasch/job_queue
#%remote_jobcatalog   <path/to/directory>
#
%local_ip            127.0.0.1
%local_username      raasch
#
%compiler_name       mpif90
%compiler_name_ser   ifort
%cpp_options         -cpp -D__parallel -DMPI_REAL=MPI_DOUBLE_PRECISION
                     -DMPI_2REAL=MPI_2DOUBLE_PRECISION -D__fftw -D__netcdf
%make_options        -j 4
%compiler_options    -fpe0 -O3 -xHost -fp-model source -ftz -fno-alias ...
                     -I /<path_to>/fftw/3.3.4/include ...
                     -L/<path_to>/fftw/3.3.4/lib64 -lfftw3 ...
%linker_options      -openmp -fpe0 -O3 -xHost -fp-model source -ftz -fno-alias ...
                     -I /<path_to>/fftw/3.3.4/include ...
                     -L/<path_to>/fftw/3.3.4/lib64 -lfftw3 ...
%hostfile            auto
%execute_command     mpiexec  -machinefile hostfile  -n {{MPI_TASKS}}  palm
#
#------------------------------------------------------------------------------
# INPUT-commands, executed before running PALM - lines must start with "IC:"
#------------------------------------------------------------------------------
#IC:
#
#------------------------------------------------------------------------------
# ERROR-commands - executed when program terminates abnormally
#------------------------------------------------------------------------------
EC:[[ \$locat = execution ]]  &&  cat  RUN_CONTROL
EC:[[ \$locat = execution ]]  &&  cat  PARTICLE_INFOS/*
#
#------------------------------------------------------------------------------
# OUTPUT-commands - executed when program terminates normally
#------------------------------------------------------------------------------
#
# Combine 1D- and 3D-profile output (these files are not usable for plotting)
OC:[[ -f LIST_PROFIL_1D     ]]  &&  cat  LIST_PROFIL_1D  >>  LIST_PROFILE
OC:[[ -f LIST_PROFIL        ]]  &&  cat  LIST_PROFIL     >>  LIST_PROFILE
#
# Combine all particle information files
OC:[[ -f PARTICLE_INFOS/_0000 ]]  &&  cat  PARTICLE_INFOS/* >> PARTICLE_INFO
```

- Created during automatic installation.
- Computer- and software-specific configuration.
- Variables are interpreted by **palmrun** script.
- **<ci>** can be an arbitrary string.
- One individually named file per environment in that PALM shall be executed (compiler options, computer, software).
- Documentation available at:

  https://palm.muk.uni-hannover.de/trac/wiki/doc/app/palm_config

Configuration of Environment – Configuration File

.palm.config.<ci>

```
#
%base_directory      /home/raasch/palm/current_version
%base_data           /home/raasch/palm/current_version/JOBS
%source_path         /home/raasch/palm/current_version/trunk/packages/palm/model/src
%user_source_path    /home/raasch/palm/current_version/JOBS/$run_identifier/USER_CODE
%fast_io_catalog     /localdata/raasch
%restart_data_path   /localdata/raasch
%output_data_path    /home/raasch/palm/current_version/JOBS
%local_jobcatalog    /home/raasch/job_queue
#%remote_jobcatalog   <path/to/directory>
#
%local_ip            127.0.0.1
%local_username      raasch
#
%compiler_name       mpiifx
%compiler_name_ser   ifx
%cpp_options         -fpp -D__parallel -DMPI_REAL=MPI_DOUBLE_PRECISION
                     -DMPI_2REAL=MPI_2DOUBLE_PRECISION -D__fftw -D__netcdf
%make_options        -j 4
%compiler_options    -fpe0 -O3 -xHost -fp-model source -ftz -fno-alias ...
                     -I /<path_to>/fftw/3.3.4/include ...
                     -L/<path_to>/fftw/3.3.4/lib64 -lfftw3 ...
%linker_options      -openmp -fpe0 -O3 -xHost -fp-model source -ftz -fno-alias ...
                     -I /<path_to>/fftw/3.3.4/include ...
                     -L/<path_to>/fftw/3.3.4/lib64 -lfftw3 ...
%hostfile            auto
%execute_command     mpiexec  -machinefile hostfile  -n {{MPI_TASKS}}  palm
#
#
# INPUT-commands, executed before running PALM - lines must start with "IC:"
#-----------------------------------------------------------------------------
#IC:
#
#-----------------------------------------------------------------------------
# ERROR-commands - executed when program terminates abnormally
#-----------------------------------------------------------------------------
EC:[[ \$locat = execution ]]  &&  cat  RUN_CONTROL
EC:[[ \$locat = execution ]]  &&  cat  PARTICLE_INFOS/*
#
#-----------------------------------------------------------------------------
# OUTPUT-commands - executed when program terminates normally
#-----------------------------------------------------------------------------
#
# Combine 1D- and 3D-profile output (these files are not usable for plotting)
OC:[[ -f LIST_PROFIL_1D     ]]  &&  cat  LIST_PROFIL_1D  >>  LIST_PROFILE
OC:[[ -f LIST_PROFIL        ]]  &&  cat  LIST_PROFIL     >>  LIST_PROFILE
#
# Combine all particle information files
OC:[[ -f PARTICLE_INFOS/_0000 ]]  &&  cat  PARTICLE_INFOS/* >> PARTICLE_INFO
```

Header

Path variables

Information about local computer

Compiler names (parallel and serial)

Preprocessor directives

Compiler and linker options, including software library paths

UNIX commands to be executed...

- before PALM code execution
- in case of an error during execution
- after PALM code execution
- **Configuration files must be adapted for batch mode and submission of jobs from a local to a remote computer!**

PALM steering

Configuration of Environment– PALM I/O configuration file

`.palm.iofiles`

```
#
#-----------------------------------------------------------------------------------------
# List of input-files
#-----------------------------------------------------------------------------------------
PARIN                      in:tr      d3#      $base_data/$run_identifier/INPUT          _p3d*
PARIN                      in:tr      d3r      $base_data/$run_identifier/INPUT          _p3dr*
PARIN                      in:tr      pcr      $base_data/$run_identifier/INPUT          _pcr*
NAVIGATION_DATA            inopt:tr   d3#:d3r  $base_data/$run_identifier/INPUT          _nav*
TOPOGRAPHY_DATA            inopt:tr   d3#:d3r  $base_data/$run_identifier/INPUT          _topo*
...
DATA_1D_PR_NETCDF          inopt      prr      $output_data_path/$run_identifier/OUTPUT  _pr*       nc
DATA_1D_TS_NETCDF          inopt      tsr      $output_data_path/$run_identifier/OUTPUT  _ts*       nc
DATA_1D_PTS_NETCDF         inopt      ptsr     $output_data_path/$run_identifier/OUTPUT  _pts*      nc
DATA_2D_XY_NETCDF          inopt      xyr      $output_data_path/$run_identifier/OUTPUT  _xy*       nc
DATA_2D_XY_NETCDF          inopt      xyr      $output_data_path/$run_identifier/OUTPUT  _xy*       nc
DATA_2D_XY_AV_NETCDF       inopt      xyr      $output_data_path/$run_identifier/OUTPUT  _av_xy*    nc
...
#
#-----------------------------------------------------------------------------------------
# List of output-files
#-----------------------------------------------------------------------------------------
BINOUT*                      out:lnpe restart   $restart_data_path/$run_identifier/RESTART  _d3d
PARTICLE_RESTART_DATA_OUT*   out:lnpe prt#:prtr $restart_data_path/$run_identifier/RESTART  _rprt
SVFOUT*                      out:lnpe svfout    $restart_data_path/$run_identifier/SVF      _svf
#
RUN_CONTROL*                 out:tr   d3#:pcr   $output_data_path/$run_identifier/MONITORING  _rc
RUN_CONTROL*                 out:tra  d3r       $output_data_path/$run_identifier/MONITORING  _rc
HEADER*                      out:tr   d3#:pcr   $output_data_path/$run_identifier/MONITORING  _header
HEADER*                      out:tra  d3r       $output_data_path/$run_identifier/MONITORING  _header
CPU_MEASURES*                out:tr   d3#:pcr   $output_data_path/$run_identifier/MONITORING  _cpu
CPU_MEASURES*                out:tra  d3r       $output_data_path/$run_identifier/MONITORING  _cpu
PARTICLE_INFO*               out:tr   pt#       $output_data_path/$run_identifier/MONITORING  _prt_info
PARTICLE_INFO*               out:tra  ptr       $output_data_path/$run_identifier/MONITORING  _prt_info
LIST_PROFIL*                 out:tr   *         $output_data_path/$run_identifier/MONITORING  _list_pr
#
DATA_1D_PR_NETCDF*           out:tr   *         $output_data_path/$run_identifier/OUTPUT  _pr        nc
DATA_1D_SP_NETCDF*           out:tr   *         $output_data_path/$run_identifier/OUTPUT  _sp        nc
DATA_1D_TS_NETCDF*           out:tr   *         $output_data_path/$run_identifier/OUTPUT  _ts        nc
DATA_1D_PTS_NETCDF           out:tr   *         $output_data_path/$run_identifier/OUTPUT  _pts       nc
DATA_2D_XY_NETCDF*           out:tr   *         $output_data_path/$run_identifier/OUTPUT  _xy        nc
DATA_2D_XY_AV_NETCDF*        out:tr   *         $output_data_path/$run_identifier/OUTPUT  _av_xy     nc
...
```

Documentation available at:
https://palm.muk.uni-hannover.de/trac/wiki/doc/app/palm_iofiles

- A default file in folder **`palm_model_system/packages/palm/model/share/config`** is used (contains a maximum of six columns)
- Definition of input/output files and paths, interpreted by **`palmrun`** script (columns 1 and 4-6)
- **Activation strings** are given with **`palmrun`** option **`-a`**
- Only one **`.palm.iofiles`** file required. Path variable **`base_data`** can be individually set in **`.palm.config.<ci>`**, and **`run_identifier`** equals the run-identifier name given with **`palmrun`** option **`-r`** (next slide)

Starting a run – palmrun options

- **`palmrun ...`**
- Important options for interactive run on local PC:
  - **`-a`** "d3#"
    (activation string, possible strings listed in **`.palm.iofiles`**)
  - **`-r`** <run_identifier>
    (e.g. run_x, taken as base file/directory name of all data files of this run)
  - **`-c`** <configuration_identifier>
    (as in configuration file name **`.palm.config.<ci>`**)
  - **`-X`** <number of processor cores (PEs) to be used>
- For batch mode and remote jobs, additional options are available, see palmrun documentation:
  - **`-b`** (run in batch mode)
  - **`-q`** <name of batch-job queue>
  - **`-t`** <CPU time (in seconds)>
  - **`-T`** <number of mpi tasks to be started on one node of the computer>

`palmrun ?` lists all available options

Starting a run – palmrun execution

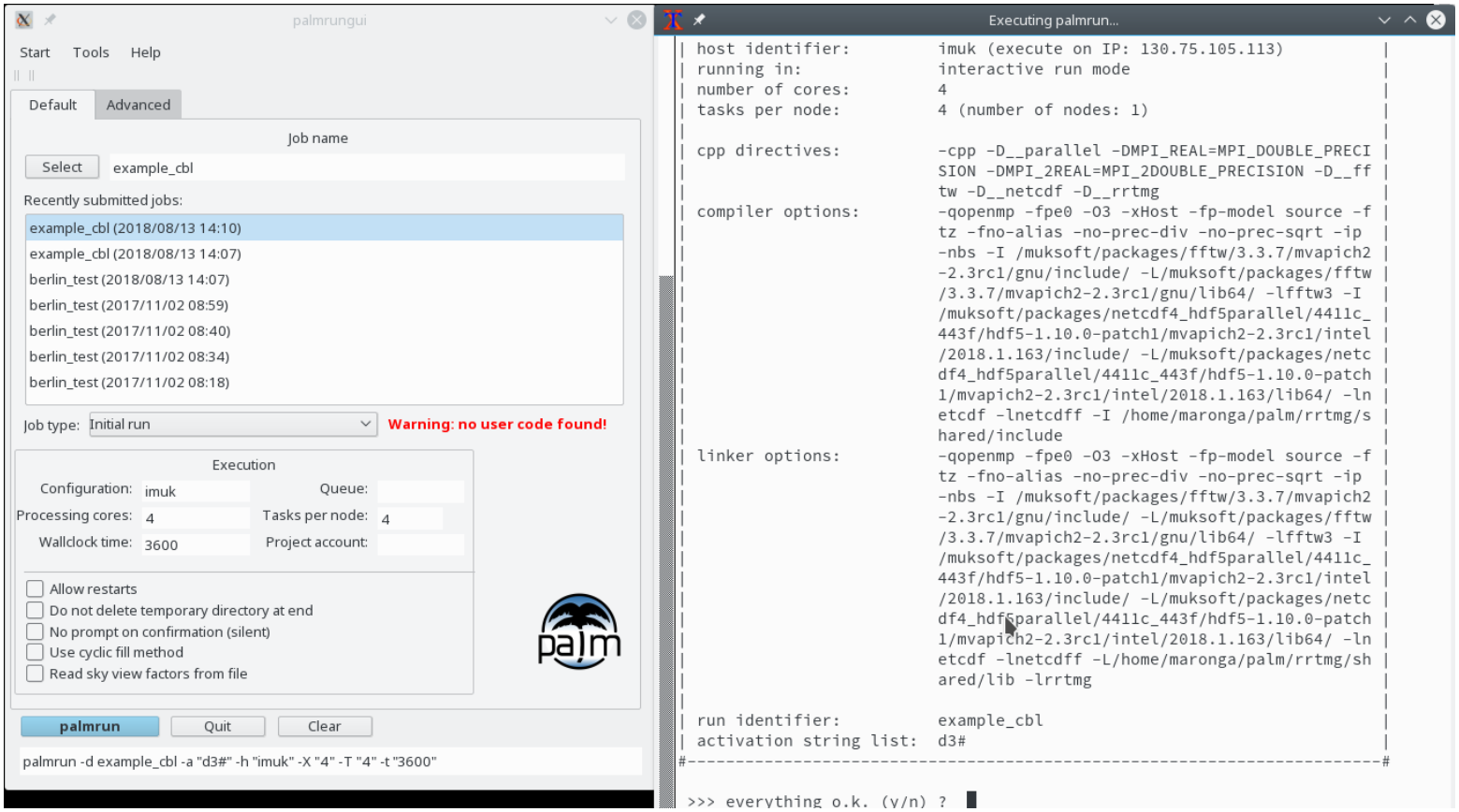
Directly type **`palmrun`** command-line into UNIX terminal

```
bora:~/palm/current_version[2018]$ palmrun -c imuk -r example_cbl -X4 -a "d3#"

*** palmrun
    will be executed.     Please wait ...

    Reading the configuration file...
    Reading the I/O files...

  *** INFORMATIVE: additional source code directory
      "/home/raasch/palm/current_version/JOBS/example_cbl/USER_CODE"
      does not exist or is not a directory.
      No source code will be used from this directory!

#--------------------------------------------------------------------------#
| palmrun                                Wed Jul 30 04:02:03 PM CEST 2025 |
| Version: PALM (git SHA-1): c19f7c52                                     |
|                                                                         |
| called on:                bora                                          |
| config. identifier:       imuk (execute on IP: 127.0.0.1)               |
| running in:               interactive run mode                          |
| number of cores:          4                                             |
| tasks per node:           4 (number of nodes: 1)                        |
|                                                                         |
| cpp directives:           -cpp -D__gfortran -D__parallel -D__netcdf -D_ |
|                           _netcdf4 -D__netcdf4_parallel -D__fftw -D__rr |
|                           tmg -D__single_precision                      |
| compiler options:         -O3 -g -ffpe-trap=invalid,zero,overflow -fbac |
|                           ...                                           |
| linker options:           -O3 -g -ffpe-trap=invalid,zero,overflow -fbac |
|                           ktrace -Wall -Wextra -Werror -Wno-unused-dumm |
|                           y-argument -Wno-do-subscript -Wno-maybe-unini |
|                           tialized -Wno-conversion -Wno-compare-reals - |
|                           Wno-surprising -pedantic -ffree-line-length-n |
|                           ...                                           |
|                                                                         |
| run identifier:           example_cbl                                   |
| activation string list:   d3#                                           |
#--------------------------------------------------------------------------#

 >>> everything o.k. (y/n) ?
```

Starting a run – palmrun execution with GUI

- Generate **`palmrun`** UNIX command-line via graphical user interface (gui)
  - (https://palm.muk.uni-hannover.de/trac/wiki/doc/app/palmrungui)
- Invoke **`palmrungui`** in UNIX terminal

```
sno:~/palm/current_version$ palmrungui
```



Summary of palmrun's workflow

Minimum command for executing PALM with more than one core:

```
palmrun -r <run_identifier> -a <activation_string>
        -X <total number of cores>
```

example: **palmrun -r *run_x* -a "*d3#*" -X *4* [-c default]**

1. Checks, if source code has been pre-compiled, i.e. does directory **MAKE_DEPOSITORY_default** exist? If not, **palmrun** calls the **palmbuild** script to compile the code according to settings in **.palm.config.default**
2. Temporary working directory **run_x.<randomnumber>** under path given by **%fast_io_catalog** in **.palm.config.default** is created
3. INPUT files and **MAKE_DEPOSITORY_default** content are copied to temporary directory. For every manual execution of **palmrun -r run_x** ..., a folder **SOURCES_FOR_RUN_run_x** is created under path **%fast_io_catalog**. It contains the compiled code, **palmrun** script, configuration files, etc.
4. PALM is executed in the temporary directory
5. MONITORING and OUTPUT files are copied from temporary directory to their destinations given in **.palm.iofiles**

```
#------------------------------------------------------------------------------
# List of output-files
#------------------------------------------------------------------------------
RUN_CONTROL*         out:tr   d3#:pcr   $base_data/$run_identifier/MONITORING  _rc
...
DATA_1D_TS_NETCDF*  out:tr         *    $base_data/$run_identifier/OUTPUT      _ts     nc
```

copy to → rename to → **run_x_ts.<cyclenumber>.nc**
where <cyclenumber> is a 3 digit number

Final filename (e.g.): $HOME/palm/current_version/JOBS/run_x/OUTPUT/run_x_ts.000.nc

6. Delete temporary directory (prevented by **palmrun** option **"-B"**).
   Folder **SOURCES_FOR_RUN_run_x** is not deleted, remains available for restart runs.

Monitoring of a run

(1) Watch the informative messages and progress bar in terminal (interactive run only).

```
palmrun -r run_x -a "d3#" -c <ci> -X 4
...
...
*** execution starts in directory
    "/localdata/palm/TMP/example_cbl.14917"
------------------------------------------------------------------------------
*** running on: mypc mypc mypc mypc
                ... reading environment parameters from ENVPAR --- finished
                ... reading NAMELIST parameters from PARIN --- finished
                ... creating virtual PE grids + MPI derived data types --- finished
                ... checking parameters --- finished
                ... allocating arrays --- finished
                ... initializing with constant profiles --- finished
                ... initializing statistics , boundary conditions , etc. --- finished
                ... creating initial disturbances --- finished
                ... calling pressure solver --- finished
                --- leaving init_3d_model
                --- start with time-stepping
[XXXXXXXXXXXXXXXXXXXXXXXXXXXXX_________________________________] 53.3 left
```

(2) Watch file RUN_CONTROL file in the temporary working directory:

- switch to temporary directory of the specific run
- command **`tail -f RUN_CONTROL`** get's you the consecutive PALM output into this file on the terminal

```
host:...$ cd <path_to_temp_directory>
host:/path_to_temp_directory$ tail -f RUN_CONTROL
0      2 00:00:40.00  20.0000A  -0.2117  -0.2252   0.1162  0.009  0.69  -0.132E+02    100.  0.102E-02  0.102E-02  0.939E-05
0.413E-15    7   7  11   11   7  19    5  26   7    0.000    0.000       0
0      3 00:01:00.00  20.0000A  -0.2106  -0.2227   0.1152  0.009  0.69  -0.132E+02    100.  0.101E-02  0.101E-02  0.131E-04
0.417E-15    7   7  11   11   7  19    5  26   7    0.000    0.000       0
```